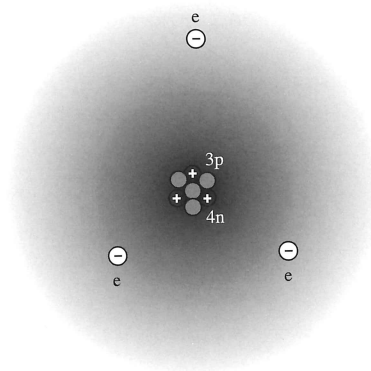


Set 8: Atoms and isotopes

Notes

Set 8: Exercises

1. a) Label the parts of the atom shown below. Note: This diagram is not to scale, the nucleus is hundreds of times smaller than shown.



- b) Complete the following using the atom shown.

Number of protons _____

Number of neutrons _____

Number of electrons _____

- c) Use a Periodic Table to identify the element. _____

2. Complete the table below.:

Particle	Charge	Mass relative to a proton
proton		
neutron		
electron		

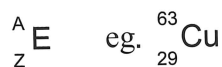
3. If an electron was removed from an atom:

- a) How would it affect the overall mass of the atom?
b) How would it affect the overall charge of the atom?

4. Copy and complete the sentence:

Isotopes of an element are atoms of the same element and hence have the same number of _____ and _____, but have different numbers of _____.

5. Isotopes are regularly represented by the following symbol :



For the element copper, identify:

E (element symbol): _____

A (mass number): _____

Z (atomic number): _____

6. Copy and complete the following table:

Symbol	Element	Mass No (A)	Atomic No (Z)	No of neutrons
$^{14}_6\text{C}$		14	6	
	chlorine		17	18
$^{56}_{26}\text{Fe}$		56	26	
		31	15	
	silver	108		61
	carbon	12		
	sodium	23	11	
$^{64}_{29}\text{Cu}$		64	29	
	calcium	40		20
	carbon			7

7. Which element in the table in question 6 is represented by more than one isotope?

8. Use the format: $^{12}_6\text{C}$ to represent the following isotopes of hydrogen.

- (a) hydrogen-1
- (b) hydrogen-2
- (c) hydrogen-3

Notes